

Exercises w.35

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Exercise 1

We want to show that:

$$\frac{\partial(\mathbf{a}^T \mathbf{x})}{d\mathbf{x}} = \mathbf{a}^T$$

to do that we define a function $f(\mathbf{x})$:

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} = \sum_{i=0}^{n-1} a_i \cdot x_i$$

then we derivative $f(\mathbf{x})$ for x_k .

$$\frac{\partial f(\mathbf{x})}{\partial x_k} = \frac{\partial}{\partial x_k} \left(\sum_{i=0}^{n-1} a_i \cdot x_i \right) = a_k$$

the derivative of the vector is written as a row vector:

$$\frac{\partial f}{\partial \mathbf{x}} = \left[\frac{\partial f(x)}{\partial x_0}, \frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_{n-1}} \right] = [a_0, a_1, \dots, a_{n-1}] = \mathbf{a}^T$$

We also want to show that:

$$\frac{\partial(\mathbf{a}^T \mathbf{A} \mathbf{a})}{d\mathbf{a}} = \mathbf{a}^T (\mathbf{A} + \mathbf{A}^T)$$

to that we define $g(\mathbf{a})$

$$g(\mathbf{a}) = \sum_{i=1}^n \sum_{j=1}^n a_i A_{ij} a_j$$

We then derivative $g(\mathbf{x})$ with x_k and get.

$$\frac{\partial g(\mathbf{a})}{\partial a_k} = \sum_{j=1}^n \frac{\partial}{\partial a_k} (a_k A_{kj} a_j) + \sum_{i=1}^n \frac{\partial}{\partial a_k} (a_i A_{ik} a_k) = \sum_{j=1}^n A_{kj} a_j + \sum_{i=1}^n a_j A_{ik}$$

on matrix form we get:

$$\frac{\partial(\mathbf{a}^T \mathbf{A} \mathbf{a})}{d\mathbf{a}} = \frac{\partial g(\mathbf{a})}{\partial \mathbf{a}} = \mathbf{a}^T \mathbf{A} + \mathbf{a}^T \mathbf{A}^T = \mathbf{a}^T (\mathbf{A} + \mathbf{A}^T)$$

Lastly we want to show that

$$\frac{\partial(\mathbf{x} - \mathbf{A}\mathbf{s})^T(\mathbf{x} - \mathbf{A}\mathbf{s})}{d\mathbf{s}} = -2(\mathbf{x} - \mathbf{A}\mathbf{s})^T \mathbf{A}$$

We start with expanding the expression.

$$(\mathbf{x} - \mathbf{A}\mathbf{s})^T(\mathbf{x} - \mathbf{A}\mathbf{s}) = \mathbf{x}^T \mathbf{x} - \mathbf{x}^T \mathbf{A}\mathbf{s} - \mathbf{s}^T \mathbf{A}^T \mathbf{x} + \mathbf{s}^T \mathbf{A}^T \mathbf{A}\mathbf{s} = \mathbf{x}^T \mathbf{x} - 2\mathbf{x}^T \mathbf{A}\mathbf{s} + \mathbf{s}^T \mathbf{A}^T \mathbf{A}\mathbf{s}$$

We will now derivative part wise and use the derivatives we showed earlier.

$$\frac{\partial(\mathbf{x}^T \mathbf{x})}{\partial \mathbf{s}} = 0$$

Since it does not include an $\mathbf{s}$ element.

in the next part we set $\mathbf{x}^T \mathbf{A}$ as $\mathbf{a}^T$ and use the first deviation

$$\frac{\partial(\mathbf{x}^T \mathbf{A}\mathbf{s})}{\partial \mathbf{s}} = \frac{\partial(\mathbf{a}^T \mathbf{s})}{\partial \mathbf{s}} = \mathbf{a}^T = (\mathbf{x}^T \mathbf{A})^T = \mathbf{A}^T \mathbf{x}$$

Lastly, we set $\mathbf{A}^T \mathbf{A}$ as a matrix $\mathbf{B}$, use the second derivative, and get

$$\begin{aligned} \frac{\partial(\mathbf{s}^T \mathbf{A}^T \mathbf{A}\mathbf{s})}{\partial \mathbf{s}} &= \frac{\partial(\mathbf{s}^T \mathbf{B}\mathbf{s})}{\partial \mathbf{s}} \\ &= \mathbf{s}^T (\mathbf{B} + \mathbf{B}^T) \\ &= \mathbf{s}^T (\mathbf{A}^T \mathbf{A} + (\mathbf{A}^T \mathbf{A})^T) \\ &= \mathbf{s}^T (\mathbf{A}^T \mathbf{A} + (\mathbf{A}^T \mathbf{A})) \\ &= 2\mathbf{s}^T \mathbf{A}^T \mathbf{A} \end{aligned}$$

we now goes back to the full equation and get

$$\begin{aligned} \frac{\partial(\mathbf{x} - \mathbf{A}\mathbf{s})^T(\mathbf{x} - \mathbf{A}\mathbf{s})}{d\mathbf{s}} &= 0 - 2\mathbf{A}^T \mathbf{x} + 2\mathbf{s}^T \mathbf{A}^T \mathbf{A} \\ &= 2(\mathbf{s}^T - \mathbf{A}^T \mathbf{A} - \mathbf{x}^T \mathbf{A}) \\ &= 2(\mathbf{s}^T \mathbf{A}^T - \mathbf{x}^T) \mathbf{A} \\ &= -2(\mathbf{x} - \mathbf{A}\mathbf{s})^T \mathbf{A} \end{aligned}$$

that was what we wanted to show.

We also want to find the second derivative of this function

$$\frac{\partial^2(\mathbf{x} - \mathbf{A}\mathbf{s})^T(\mathbf{x} - \mathbf{A}\mathbf{s})}{d\mathbf{s}^2} = \frac{\partial(-2(\mathbf{x} - \mathbf{A}\mathbf{s})^T\mathbf{A})}{d\mathbf{s}} = \frac{\partial(-2\mathbf{A}^T\mathbf{x} + 2\mathbf{s}^T\mathbf{A}^T\mathbf{A})}{d\mathbf{s}}$$

we will now do part wise derivation

$$\frac{\partial(-2\mathbf{A}^T\mathbf{x})}{d\mathbf{s}} = 0$$

since the equation does not include $\mathbf{s}$

we define a vector $h(\mathbf{s})$ for the second element, and $\mathbf{A}^T\mathbf{A}$ as $\mathbf{B}$

$$h(\mathbf{s}) = 2\mathbf{s}^T\mathbf{A}^T\mathbf{A} = 2\mathbf{s}^T\mathbf{B}$$

We have a vector so we look at the k-th element.

$$h_k(\mathbf{s}) = (2\mathbf{s}^T\mathbf{B})_k = 2 \sum_{i=1}^n s_i B_{ik} = 2\mathbf{s}^T\mathbf{b}_k$$

and derivative $h_k(\mathbf{s})$

$$\frac{\partial h_k(\mathbf{s})}{\partial s} = (2\mathbf{b}_k)^T$$

that gives us

$$\frac{\partial h(\mathbf{s})}{\partial s} = 2\mathbf{B}^T = 2(\mathbf{A}^T\mathbf{A})^T = 2\mathbf{A}^T\mathbf{A}$$

so the second derivative of the function is

$$\frac{\partial^2(\mathbf{x} - \mathbf{A}\mathbf{s})^T(\mathbf{x} - \mathbf{A}\mathbf{s})}{d\mathbf{s}^2} = 2\mathbf{A}^T\mathbf{A}$$